Dual-Axis Solar Panel Gimbal Mechanism

Mechanical Design Assessment – Digantara

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1. Requirements Definition

1.1 Input Specifications

Parameter	Specification
Solar Panel Mass	2 kg
Solar Panel Size	400 × 200 × 10 mm
Pitch Rotation	±45°
Yaw Rotation	±45°
Independent Motion	Yes

1.2 Program Constraints

The conceptual mechanism shall satisfy the following project constraints:

- Budget-constrained development.
- First launch targeted within 15 months.
- Production readiness within 18–24 months.
- Low development risk.
- High reliability.
- Scalable for future production.

1.3 Functional Requirements

The proposed mechanism shall:

- Support a 2 kg deployable solar panel.
- Provide independent pitch and yaw motion.
- Rotate ±45° about each axis.
- Maintain structural rigidity.
- Minimize overall system weight.
- Achieve accurate sun tracking.
- Enable straightforward manufacturing and assembly.
- Support future qualification and production.

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2. Design Objectives

The conceptual design has been developed considering the following engineering objectives.

Primary Objectives

- Lightweight structure
- High structural stiffness
- Compact packaging
- High positioning accuracy
- Low backlash
- Modular architecture
- Ease of manufacturing
- Ease of maintenance
- Cost-effective production
- High reliability

Secondary Objectives

- Reduced part count
- Easy actuator replacement
- Future scalability
- Ease of qualification testing

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3. Design Assumptions

The following assumptions have been considered during conceptual development.

- The spacecraft team supplies a completely manufactured solar panel.
- Solar panel dimensions are fixed at **400 × 200 × 10 mm**.
- Solar panel mass is **2 kg**.
- Structural mounting inserts are already available.
- Solar panel deployment has already occurred.
- Continuous 360° rotation is not required.
- Maximum required motion is limited to ±45°.
- Flexible electrical harnesses accommodate rotational movement.
- Slip rings are not required.
- Space-qualified bearings and actuators are commercially available.
- Internal gearbox design is outside the scope of this conceptual assessment.
- Preliminary CAD represents the external mechanical architecture only.

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4. Concept Selection

After evaluating multiple design concepts, a **Dual-Axis Gimbal Architecture** was selected.

Configuration

- Independent Pitch Axis
- Independent Yaw Axis
- Compact U-frame
- Direct actuator mounting
- Modular construction

Alternative Concepts Considered

Concept 1

Servo-driven universal joint

Advantages

- Compact

Disadvantages

- Difficult assembly
- Complex packaging
- Difficult maintenance

Concept 2

Belt-driven dual-axis system

Advantages

- Lower motor placement

Disadvantages

- Belt stretch
- Lower positioning accuracy
- Increased maintenance

Concept 3 (Selected)

Independent Dual-Axis Gimbal

Advantages

- Simple mechanism
- Excellent structural stiffness
- Easy assembly
- Easy maintenance
- Aerospace-proven architecture
- Low development risk

This concept was therefore selected

5. System Architecture

Solar Panel
|
Mounting Lugs
|
Pitch Shaft
|
Pitch Bearings
|
Pitch Actuator
|
Lightweight U-Frame
|
Yaw Shaft
|
Yaw Bearings
|
Yaw Actuator
|
Base Plate
|
Spacecraft Structure

Power Flow

Pitch Axis

BLDC Servo Motor

↓

Harmonic Drive

↓

Pitch Shaft

↓

Solar Panel

Yaw Axis

BLDC Servo Motor

↓

Harmonic Drive

↓

Yaw Shaft

↓

Complete Pitch Assembly

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6. Working Principle

The proposed mechanism consists of two independent rotational axes.

Pitch Motion

The pitch actuator is mounted directly on the U-frame.

When commanded by the spacecraft control system, the BLDC motor rotates through a harmonic drive gearbox.

The gearbox increases output torque while significantly reducing backlash.

The gearbox output is connected to the pitch shaft.

The shaft rotates the solar panel through an angle of **±45°**.

Yaw Motion

The yaw actuator is mounted beneath the U-frame.

The actuator rotates the complete pitch assembly around the vertical axis.

The solar panel therefore obtains independent yaw motion without affecting pitch control.

Maximum yaw travel is **±45°**.

Advantages

- Independent control
- Smooth motion
- High pointing accuracy
- Compact design
- Easy maintenance
- Reduced mechanical complexity

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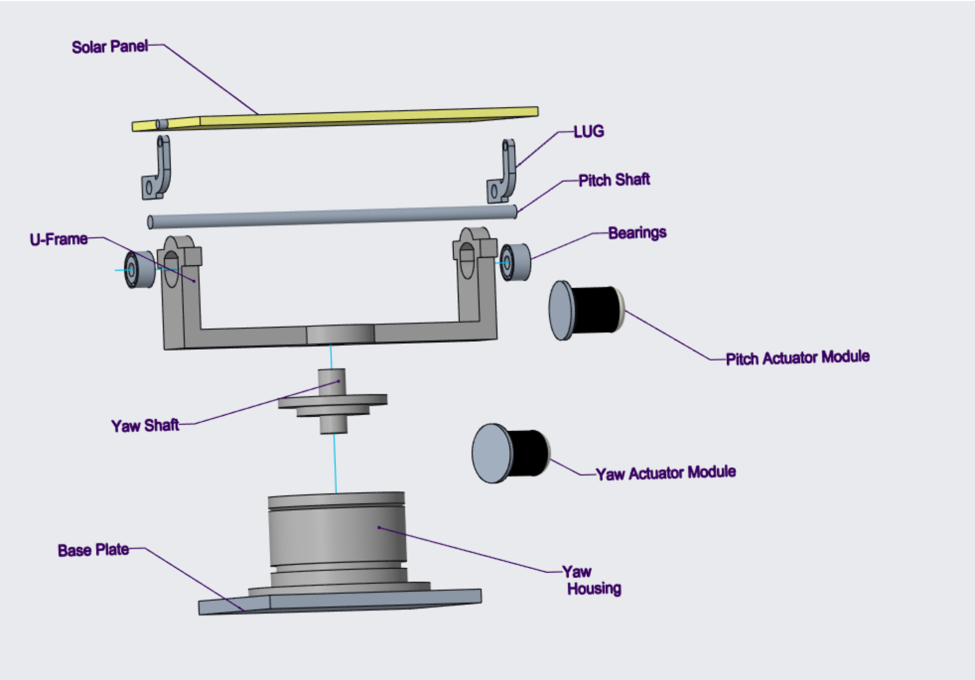
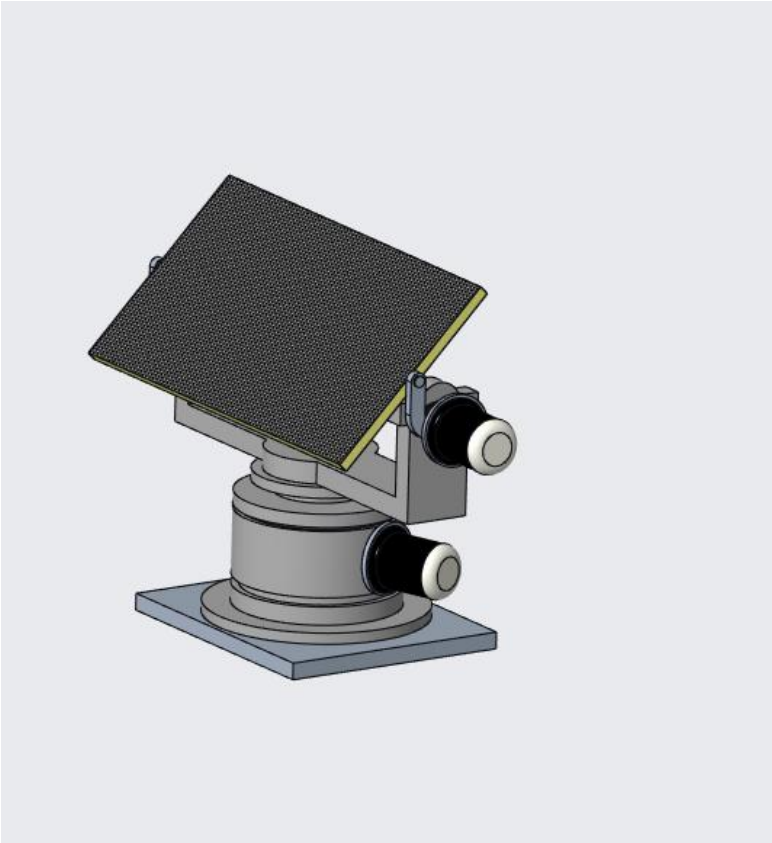
7. Preliminary CAD Model

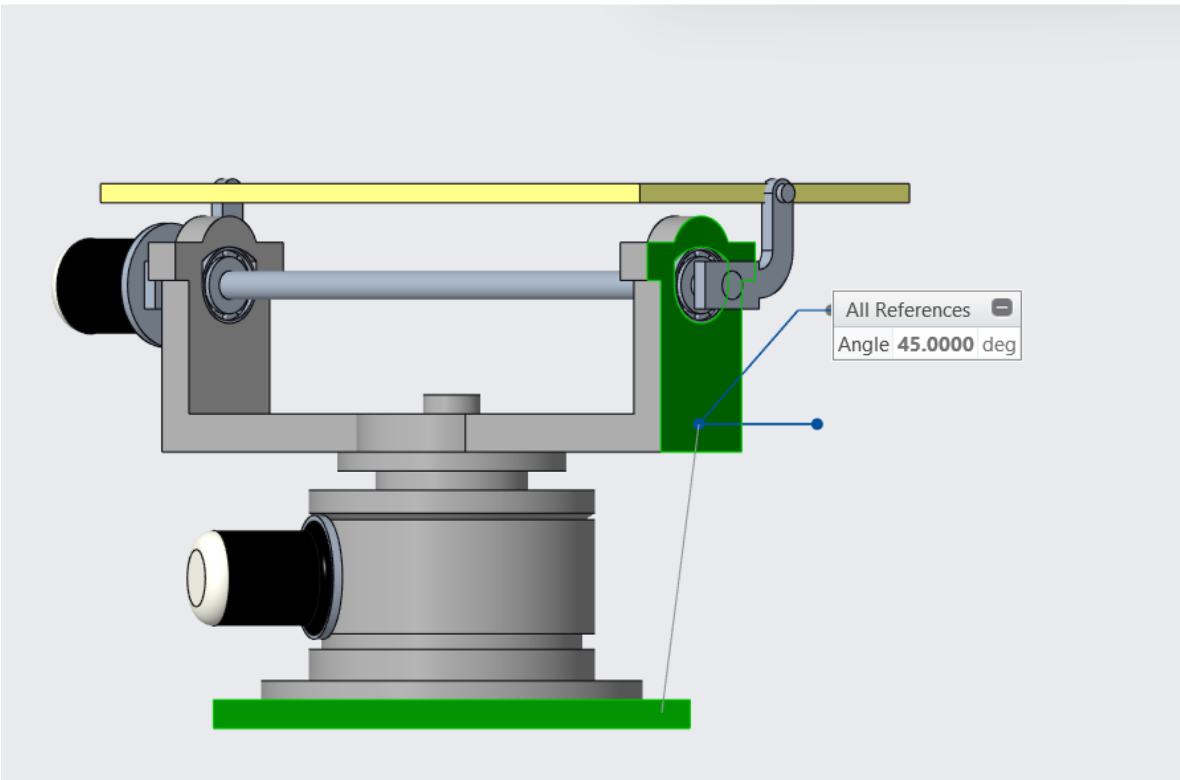
The CAD model represents the conceptual mechanical arrangement of the proposed Solar Array Drive Assembly.

The CAD model includes:

- Solar panel
- Mounting lugs
- Pitch shaft
- Pitch bearings
- Lightweight aluminium U-frame
- Pitch actuator housing
- Yaw actuator housing
- Base plate



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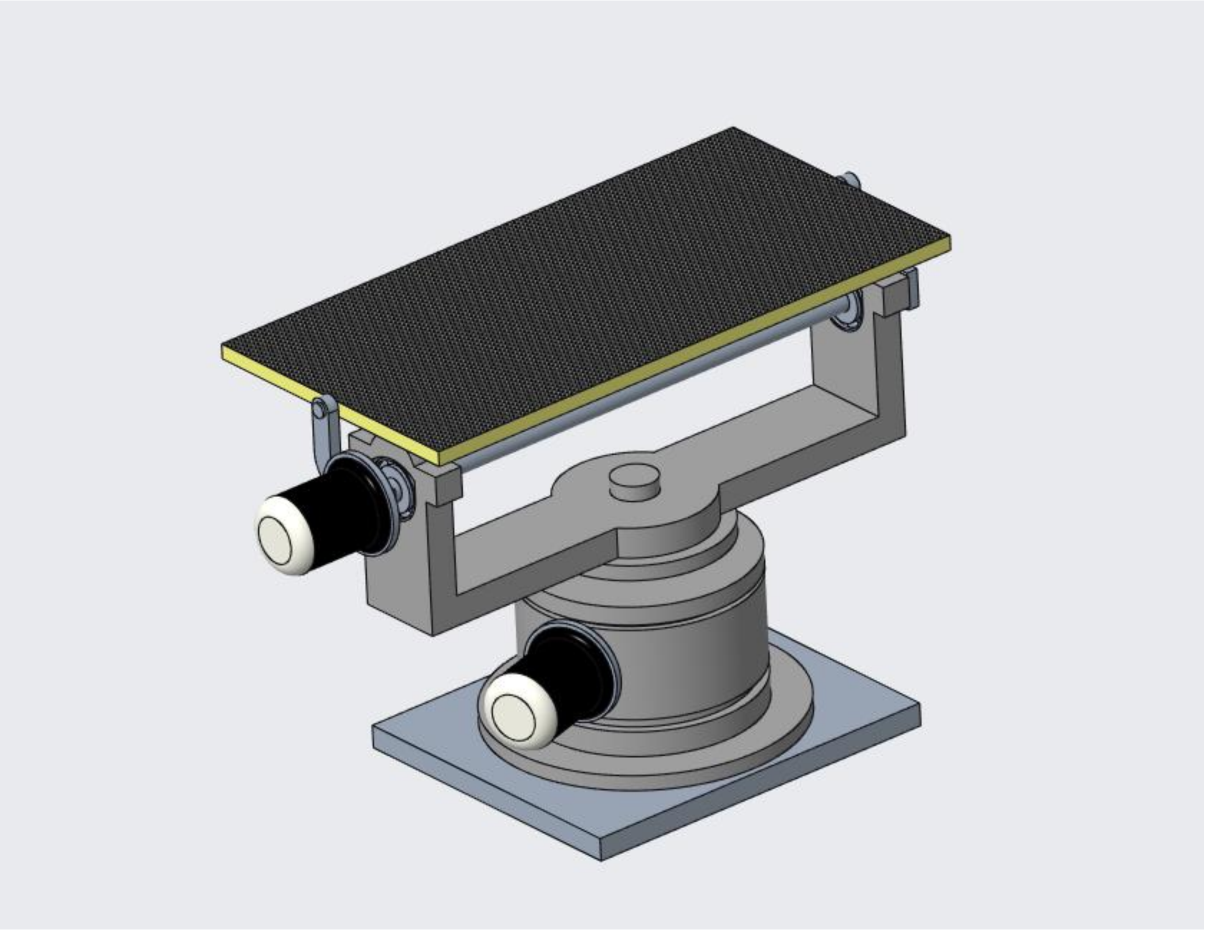

The CAD model focuses on validating the overall architecture, actuator arrangement, packaging, assembly feasibility, and mechanical motion.

Detailed internal gearbox geometry, actuator internals, bearing preload, and fastener sizing are intentionally excluded since this submission represents the conceptual design phase.

8. Actuator Selection

8.1 Selected Actuator

The proposed design utilizes **Integrated Brushless DC (BLDC) Servo Motors coupled with Harmonic Drive Gear Reduction** for both the Pitch and Yaw axes.

This actuator configuration is widely adopted in aerospace pointing mechanisms due to its excellent positioning accuracy, compact size, high torque density, and minimal backlash.

The harmonic drive gearbox significantly increases output torque while maintaining high positional precision, making it suitable for spacecraft solar array tracking applications.

8.2 Selection Criteria

Requirement	Engineering Justification
High positional accuracy	Required for precise solar tracking throughout the mission.
High torque density	Capable of supporting and rotating the 2 kg solar panel.
Low backlash	Harmonic drive minimizes positioning errors and improves pointing accuracy.
Compact packaging	Easily integrates within the spacecraft envelope.
High efficiency	Reduces electrical power consumption.
Modular design	Simplifies maintenance and actuator replacement.
Aerospace suitability	Proven technology in satellite pointing mechanisms and gimbal systems.

8.3 Why BLDC Motor?

The BLDC motor was selected due to the following advantages:

- High efficiency
- High torque-to-weight ratio
- Long operational life
- Low maintenance
- Excellent speed control
- High reliability
- Suitable for closed-loop control systems
- Compatible with spacecraft pointing applications

8.4 Why Harmonic Drive?

A harmonic drive gearbox offers several advantages compared to conventional planetary gear systems.

Benefits

- Extremely low backlash
- High reduction ratio
- Compact packaging
- High torque transmission
- High positioning accuracy
- Lightweight construction
- Proven space heritage

These characteristics make harmonic drives highly suitable for Solar Array Drive Assemblies (SADA).

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9. Mechanism Components

The conceptual mechanism consists of the following major components.

9.1 Solar Panel

The solar panel serves as the payload supported by the gimbal mechanism.

Specifications:

- Mass: 2 kg
- Size: 400 × 200 × 10 mm
- Independent Pitch and Yaw motion

9.2 Mounting Lugs

Two lightweight aluminium mounting lugs connect the solar panel to the pitch shaft.

Functions:

- Load transfer
- Structural support
- Easy assembly
- Easy replacement
- Uniform load distribution

9.3 Pitch Shaft

The pitch shaft transfers rotational torque from the actuator to the solar panel.

Functions:

- Rotational support
- Torque transmission
- Structural stiffness
- Alignment accuracy

9.4 Pitch Bearings

Angular contact bearings support the pitch shaft.

Functions:

- Smooth rotation
- Reduced friction
- High stiffness
- Improved positional accuracy

9.5 Lightweight U-Frame

The U-frame supports the entire pitch assembly.

Functions:

- Structural support
- Actuator mounting
- Bearing housing
- Load transfer
- Weight reduction

9.6 Pitch Actuator Module

Includes:

- BLDC Servo Motor
- Harmonic Drive Gearbox

- Encoder
- Housing

9.7 Yaw Housing

The yaw housing supports the complete rotating assembly.

Functions:

- Bearing support
- Actuator integration
- Structural rigidity

9.8 Yaw Shaft

The yaw shaft transmits torque to rotate the entire pitch assembly.

Functions:

- Torque transmission
- Structural support
- Rotational alignment

9.9 Yaw Bearings

The yaw bearings support the rotating pitch assembly while maintaining smooth rotational movement.

9.10 Base Plate

The base plate interfaces directly with the spacecraft primary structure.

Functions:

- Structural mounting
- Load transfer
- Actuator support
- Assembly reference

9.11 Fasteners

Standard aerospace fasteners are used throughout the assembly.

Benefits:

- Easy maintenance
- Reliable assembly
- Weight optimization
- High strength

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10. Solar Array Drive Assembly (SADA) Interface

The solar panel interfaces with the Solar Array Drive Assembly using two precision-machined aluminium mounting lugs.

The mounting lugs are bolted directly to the structural inserts provided on the solar panel.

The interface transfers structural loads efficiently while minimizing localized stress concentrations.

The mounting arrangement also allows quick replacement of the solar panel during assembly or maintenance.

Advantages

- Lightweight interface
- Uniform load distribution
- Easy assembly
- Easy replacement
- Reduced stress concentration
- High structural stiffness

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11. Tribology Considerations

The proposed mechanism operates in a vacuum environment where conventional lubricants cannot be used.

Therefore, several tribological considerations have been incorporated into the conceptual design.

Bearing Selection

Vacuum-compatible angular contact bearings are recommended.

Lubrication

Dry lubrication using **Molybdenum Disulfide (MoS_2)** coating is recommended.

Advantages include:

- Vacuum compatibility
- Low friction
- High wear resistance
- Long operational life

Backlash Control

Low backlash is essential to maintain accurate solar pointing.

The harmonic drive gearbox significantly minimizes backlash compared to conventional gear systems.

Wear Protection

Critical contact surfaces shall be surface treated to improve wear resistance and reduce friction.

Thermal Considerations

Material combinations should minimize differential thermal expansion and prevent galling during temperature cycling.

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12. Material Selection

Material selection has been based on structural strength, weight optimization, manufacturability, corrosion resistance, and aerospace suitability.

Component	Material	Justification
U-Frame	Aluminium 7075-T6	High strength-to-weight ratio
Base Plate	Aluminium 6061-T6	Easy machining and lightweight
Mounting Lugs	Aluminium 7075-T6	High stiffness
Pitch Shaft	Titanium Grade 5	High strength and reduced mass
Yaw Shaft	Titanium Grade 5	Excellent fatigue resistance
Bearings	Stainless Steel 440C	Wear resistance and hardness
Fasteners	Titanium Alloy	Lightweight and corrosion resistant

Material Selection Philosophy

The selected materials provide:

- High structural stiffness
- Low mass
- Excellent fatigue resistance
- Good machinability
- High corrosion resistance
- Long operational life.

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13. Bill of Materials

Item	Quantity
Solar Panel	1
Mounting Lugs	2
Pitch Shaft	1
Pitch Bearings	2
Lightweight U-Frame	1
Pitch Actuator	1
Yaw Housing	1
Yaw Shaft	1
Yaw Bearings	2
Yaw Actuator	1
Base Plate	1
Aerospace Fasteners	As Required

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14. Preliminary Engineering Calculations

The submitted design represents a conceptual engineering solution. Therefore, preliminary calculations have been performed to validate the feasibility of the proposed architecture.

14.1 Solar Panel Load

Solar Panel Mass = **2 kg**

Gravitational Acceleration = **9.81 m/s^2**

Applied Load:

$F = m \times g$

$F = 2 \times 9.81 = 19.62$ N

The structure is therefore designed to safely withstand the equivalent static load during handling, testing, and launch.

14.2 Preliminary Pitch Torque

Assuming the centre of gravity of the solar panel is located approximately **100 mm** from the pitch axis:

Torque:

$T = \text{Force} \times \text{Distance}$

$T = 19.62 \times 0.1$

$T = 1.96$ N·m

Considering launch dynamics and design safety margins, an actuator capable of delivering approximately **6–10 N·m** output torque after gear reduction is recommended.

14.3 Bearing Considerations

The selected bearings shall withstand:

- Radial loads
- Axial loads
- Launch vibration
- Thermal expansion
- Operational rotation

Angular contact bearings are recommended due to their superior stiffness and ability to support combined loading.

14.4 Factor of Safety

A preliminary **Factor of Safety (FoS) of 2–3** has been considered appropriate for conceptual development.

The final safety factor will be verified during detailed structural analysis and qualification testing.

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15. Design Validation Strategy

The proposed mechanism shall undergo a structured engineering validation process before qualification for spacecraft integration.

15.1 Static Structural Analysis

Static structural analysis will be performed to evaluate:

- Maximum von Mises stress
- Overall deformation
- Shaft stress
- U-frame stiffness
- Mounting lug stress
- Bearing housing deformation

The objective is to ensure that all stresses remain below the allowable material limits while maintaining adequate structural stiffness.

15.2 Modal Analysis

Modal analysis will be conducted to determine the natural frequencies of the complete assembly.

Objectives include:

- Avoid structural resonance
- Improve vibration resistance
- Ensure launch survivability
- Increase pointing stability

15.3 Launch Load Analysis

The assembly shall be verified against expected launch loading conditions.

The analysis will evaluate:

- Axial loading
- Lateral loading
- Shock loading
- Random vibration
- Structural integrity of fasteners

15.4 Thermal Assessment

Thermal behaviour shall be evaluated to study:

- Thermal expansion
- Differential material expansion
- Bearing clearance variation
- Structural distortion

15.5 Functional Testing

The following functional tests are proposed.

- Pitch motion verification
- Yaw motion verification
- Encoder accuracy
- Position repeatability
- Motion smoothness

- Backlash measurement
- End-stop verification

15.6 Environmental Qualification

Future qualification testing may include:

- Thermal Vacuum Test (TVAC)
- Random Vibration Test
- Shock Test
- Functional Cycling
- Life Cycle Testing
- Outgassing Verification

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16. GD&T and Tolerance Philosophy

High pointing accuracy requires precise control of critical interfaces.

The following GD&T strategy is recommended.

Critical Features

- Pitch shaft axis
- Yaw shaft axis
- Bearing seats
- Mounting lug interfaces
- Base plate mounting surface
- Actuator mounting interfaces

Recommended GD&T Controls

Feature	GD&T Control
Bearing Bore	Position
Shaft	Circular Runout
Mounting Face	Flatness
Actuator Mount	Position
Shaft Axis	Perpendicularity
Mounting Interface	Parallelism
Solar Panel Interface	Profile of Surface

Tolerance Philosophy

The design philosophy emphasizes:

- Accurate shaft alignment
- Reduced backlash
- Controlled bearing preload
- Improved assembly repeatability
- Simplified manufacturing
- Consistent pointing accuracy

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17. Design for Manufacturing and Assembly (DFMA)

The conceptual design incorporates Design for Manufacturing and Assembly (DFMA) principles to reduce manufacturing cost, simplify assembly, and improve maintainability.

Manufacturing Considerations

Most of structural components are designed for CNC machining using standard aerospace aluminium alloys.

Advantages include:

- High dimensional accuracy
- Excellent surface finish
- Reduced lead time
- Lower production risk
- Ease of inspection

Assembly Considerations

The mechanism has been divided into independent modules.

Modules include:

- Pitch Module
- Yaw Module
- Base Module
- Solar Panel Mounting Module

This modular approach enables easy assembly, maintenance, and future upgrades.

Maintenance Considerations

The design allows:

- Easy actuator replacement
- Easy bearing replacement
- Simplified inspection
- Reduced maintenance time
- Lower life-cycle cost

Production Readiness

The proposed architecture minimizes manufacturing complexity through:

- Standard fasteners
- Reduced part count
- Symmetrical components
- Simple machining operations
- Modular assembly philosophy

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18. Risk Assessment

Risk	Possible Impact	Mitigation Strategy
Launch vibration	Structural failure	High stiffness design with secure fasteners
Gear backlash	Reduced pointing accuracy	Harmonic drive gearbox
Bearing wear	Increased friction	Vacuum-compatible bearings with dry lubrication
Cable fatigue	Electrical failure	Flexible harness routing within ±45° rotation
Thermal expansion	Misalignment	Compatible material selection
Manufacturing delays	Schedule impact	Modular CNC-manufactured components
Assembly errors	Reduced reliability	Simple modular architecture
Weight growth	Reduced spacecraft efficiency	Lightweight material selection

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19. Future Development

The current submission represents a conceptual engineering design.

The next phase of development will include:

- Detailed actuator sizing
- Complete torque calculations
- Shaft diameter optimization
- Bearing life calculations
- Structural FEA
- Modal analysis
- Thermal analysis
- Launch load verification
- Tolerance stack-up analysis
- Cable routing optimization
- Prototype manufacturing
- Functional testing
- Qualification testing
- Flight hardware development

These activities will further improve design maturity and prepare the mechanism for spacecraft integration.

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20. Conclusion

A conceptual dual-axis Solar Panel Gimbal Mechanism has been successfully developed to satisfy the mission requirements specified in this assessment.

The proposed architecture supports a **2 kg solar panel** while providing independent **±45° pitch and yaw motion** through a compact, lightweight, and modular dual-axis gimbal configuration.

The design emphasizes high structural stiffness, manufacturability, ease of assembly, low maintenance, and future production readiness. The selection of BLDC servo motors combined with harmonic drive gearboxes ensures precise positioning, high torque capability, and minimal backlash, making the mechanism suitable for spacecraft sun-tracking applications.

Appropriate engineering considerations have been incorporated, including material selection, tribology, structural validation planning, GD&T strategy, DFMA principles, and preliminary engineering calculations. Although this submission represents a conceptual design, it establishes a strong engineering foundation for future detailed design, actuator sizing, finite element analysis, tolerance optimization, environmental qualification, and prototype validation.

The proposed concept effectively balances performance, reliability, manufacturability, and project schedule while meeting the functional objectives of the assessment. It provides a practical and scalable solution that can be further refined into a flight-qualified Solar Array Drive Assembly (SADA) for future spacecraft applications.